CF06: Complete Formalism — Information Geometry Foundations for the M-Limb (Fisher and Ruppeiner)

Justin K. Lietz
Neuroca, Inc.
justin@neuroca.ai

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Abstract

The metriplectic (GENERIC) split used by the Void Dynamics Model (VDM) requires a symmetric positive semidefinite "metric limb" that drives entropy increase without violating invariants. This CF supplies the missing information-geometric foundation for that metric limb by deriving (i) the Fisher information metric as the intrinsic quadratic form controlling local distinguishability of probability models, (ii) the Cramér–Rao bound linking Fisher geometry to achievable estimator precision, and (iii) the Ruppeiner thermodynamic metric as the (negative) entropy Hessian governing equilibrium fluctuations. We then give a canonical bridge between Fisher and Ruppeiner geometries in exponential-family/canonical ensembles via MaxEnt/Legendre duality, and we show how a Fisher/Ruppeiner metric induces a metriplectic-style dissipation bracket as a natural-gradient flow on a coarse-grained statistical manifold. Finally, we define a falsifiable gate suite for CFN implementation: coordinate-invariant curvature checks, Fisher–Ruppeiner consistency in canonical toy systems, and curvature–response diagnostics that become experimentally meaningful once a specific coarse-graining map and observable set are fixed.

Keywords information geometry · Fisher information · Cramér–Rao bound · thermodynamic geometry · Ruppeiner metric · natural gradient · metriplectic · GENERIC · entropy production · Void Dynamics Model

Contents

1 Objective and scope

This CF establishes the *information-geometric* foundation required to treat the VDM metric limb (the dissipative bracket) as more than an ad-hoc "positive operator." The goal is to make the statement

"dissipation is an epistemic projection"

precise, falsifiable, and implementable. In particular, we derive the Fisher information metric, the Cramér–Rao bound, and the Ruppeiner thermodynamic metric, and we provide an explicit bridge between Fisher and Ruppeiner geometries in canonical/exponential-family ensembles.

Canon alignment. This document follows the VDM canon conventions: axioms and gates [Lietz, 2025a], equation registry anchors [Lietz, 2025b], validation metrics [Lietz, 2025c], symbols [Lietz, 2025d], tier standards [Lietz, 2025e], and provenance conventions [Lietz, 2025f].

Scope classification. This CF is **derived-limit** until a specific identification of the metric bracket operator $M(q)$ with an information-geometric metric is fixed for a concrete model. Once such an identification is made, the gates in §2 become **axiom-relevant**: failure means the proposed "metric-as-information" identification is wrong, not merely aesthetically unappealing.

Why this matters for falsifiability. Metriplectic (GENERIC) evolution requires a symmetric positive semidefinite operator M that (i) produces nonnegative entropy production and (ii) respects degeneracy constraints so invariants are not dissipated [Morrison, 1986, Grmela and "Ottinger, 1997, "Ottinger and Grmela, 1997, "Ottinger, 2005]. Information geometry supplies a canonical source of symmetric PSD forms (Fisher/Ruppeiner) and a clear operational meaning (estimation precision and fluctuation covariances). The resulting proposal is testable: it predicts quantitative links between coarse-grained fluctuations, response functions, and geometric invariants.

2 Claim inventory and decisive falsifiers

Primary claims

Claim: C1: Fisher metric controls local distinguishability *Decisive metric: KL second-order residual* $\epsilon_{\mathrm{KL}} \leq 10^{-10}$.

For a regular parametric statistical model $p(x \mid \theta)$, the Kullback–Leibler divergence admits the expansion

$$D_{\mathrm{KL}}\big(p(\cdot \mid \theta) \,\|\, p(\cdot \mid \theta + \mathrm{d}\theta)\big) = \tfrac{1}{2}\, g^{\mathrm{F}}_{ij}(\theta)\, \mathrm{d}\theta^i\, \mathrm{d}\theta^j + \mathcal{O}(\|\mathrm{d}\theta\|^3), \tag{1}$$

where g^{F} is the Fisher information metric. This identifies Fisher geometry as the intrinsic local quadratic form governing statistical distinguishability [Amari and Nagaoka, 2000, Cover and Thomas, 2006].

Claim: C2: Fisher metric is canonical under coarse-graining *Decisive metric: monotonicity residual under Markov maps* $\epsilon_{\mathrm{mono}} \leq 10^{-10}$.

Under standard regularity assumptions, the Fisher metric is the unique (up to scale) Riemannian metric on statistical models that is monotone under stochastic (Markov) maps (Čencov/Chentsov theorem) [Chentsov, 1982, Amari, 2016].

Claim: C3: Ruppeiner metric equals fluctuation Hessian *Decisive metric: equilibrium fluctuation residual* $\epsilon_{\mathrm{fluct}} \leq 10^{-8}$.

For an equilibrium thermodynamic state space with extensive coordinates $X = (X^1, \ldots, X^n)$ and entropy $S(X)$, the Ruppeiner metric

$$g^{\mathrm{R}}_{ij}(X) := -\frac{\partial^2 (S/k_B)}{\partial X^i\, \partial X^j} \tag{2}$$

is positive semidefinite (entropy concavity) and controls Gaussian fluctuations via $\mathrm{cov}(X) \approx (g^{\mathrm{R}})^{-1}$ in the appropriate ensemble [Ruppeiner, 1979, 1995, Callen, 1985].

Claim: C4: Fisher and Ruppeiner metrics are dual in canonical ensembles *Decisive metric: relative mismatch $\epsilon_{\mathrm{FR}} \leq 10^{-8}$ on toy systems.*

For exponential-family/canonical ensembles obtained by MaxEnt with Lagrange multipliers θ, Fisher geometry on multiplier space is the covariance of sufficient statistics, while Ruppeiner geometry on expectation space is (up to coordinate conventions and the choice of dimensionless entropy) the inverse covariance. This yields an explicit Fisher–Ruppeiner bridge that is testable by simulation or experiment [Amari and Nagaoka, 2000, Jaynes, 1957a,b, Ruppeiner, 1995].

Assumption ledger

- **A1 (Regularity).** Differentiation under the integral is valid; score has finite second moments (standard in Fisher theory) [Fisher, 1925, Cramér, 1946].
- **A2 (Local model).** The parameterization is a local chart on a manifold of distributions; reparameterizations are smooth.
- **A3 (Equilibrium baseline).** Ruppeiner geometry statements are restricted to equilibrium thermodynamics unless nonequilibrium corrections are explicitly modeled [Callen, 1985, Ruppeiner, 1995].
- **A4 (No critical exponents baked in).** Any critical scaling is treated as an inferred fit parameter (or an external benchmark for comparison), not assumed.

Decisive falsifiers / gates

Gate: G1: Coordinate invariance of curvature scalars *Threshold: two-chart agreement $\leq 10^{-8}$ relative.*

Compute a scalar invariant of the metric (e.g., Ricci scalar R where defined) in two coordinate charts related by a smooth reparameterization. Require $\max |R - R'|/\max(|R|, 1) \leq 10^{-8}$. This is a direct falsifier for incorrect tensor implementation in CFN and a sanity check for numerical differentiation.

Gate: G2: Fisher–Ruppeiner bridge in canonical toy systems *Threshold: relative mismatch $\epsilon_{\mathrm{FR}} \leq 10^{-8}$.*

On canonical ensembles where both sides can be computed independently (sampling for Fisher covariance; thermodynamic derivatives for Ruppeiner Hessian), require agreement with the predicted duality relations (§6).

Gate: G3: Curvature–response diagnostics *Threshold: model-comparison score favors the geometry-linked predictor.*

When the metric is tied to a specific coarse-graining and an observable response function (heat capacity, susceptibility, transport coefficient), compare: (i) a geometry-linked predictor class (curvature/metric-derived) versus (ii) a baseline predictor class that ignores geometry. A failure of the geometry-linked class to outperform baseline (after controlling for complexity) falsifies the claimed predictive role.

CFN outputs

A companion CFN notebook for this module should emit at minimum:

- symbolic derivations for at least one exponential-family toy model (e.g., Gaussian, Bernoulli/logit),
- numeric Fisher matrices and Cramér–Rao checks (Monte Carlo),
- Ruppeiner metric computation for at least one equilibrium model (ideal gas baseline; interacting model optional),
- curvature invariants and coordinate-invariance gate logs (G1),
- Fisher–Ruppeiner cross-check plots and residuals (G2),
- provenance fields: commit, seeds, grids/step sizes, and artifact hashes.

3 Fisher information geometry

3.1 Statistical manifolds and the score

Let $\mathcal{X}$ be a sample space with reference measure $\mathrm{d}x$. A statistical model is a family $\{p(x \mid \theta) : \theta \in \Theta\}$ where $\Theta \subset \mathbb{R}^n$ is open and $p(x \mid \theta) \geq 0$, $\int p(x \mid \theta)\,\mathrm{d}x = 1$. Define the *score* (log-derivative)

$$\ell_i(x;\theta) := \partial_i \log p(x \mid \theta). \tag{3}$$

Under A1, the score is centered:

$$\mathbb{E}_\theta[\ell_i] = \int p\,\partial_i \log p\,\mathrm{d}x = \int \partial_i p\,\mathrm{d}x = \partial_i \int p\,\mathrm{d}x = 0. \tag{4}$$

3.2 Definition of the Fisher information metric

Definition 1 (Fisher information matrix / Fisher–Rao metric)**.** The Fisher information matrix is

$$g^{\mathrm{F}}_{ij}(\theta) := \mathbb{E}_\theta\big[\ell_i\,\ell_j\big] = \int p(x \mid \theta)\,\partial_i \log p(x \mid \theta)\,\partial_j \log p(x \mid \theta)\,\mathrm{d}x. \tag{5}$$

Equivalently, under A1 one may use

$$g^{\mathrm{F}}_{ij}(\theta) = -\mathbb{E}_\theta\big[\partial_i\partial_j \log p(x \mid \theta)\big]. \tag{6}$$

The Fisher metric is symmetric by construction and positive semidefinite because it is an expectation of a square:

$$v^i g^{\mathrm{F}}_{ij} v^j = \mathbb{E}_\theta\Big[\big(v^i \ell_i\big)^2\Big] \geq 0. \tag{7}$$

It is positive definite on directions that change the distribution (non-identifiable directions form the nullspace).

3.3 Derivation from local KL divergence

A key point for VDM is that Fisher geometry is not a stylistic choice: it appears as the local quadratic approximation to the KL divergence. Let $D_{\mathrm{KL}}(p\|q) = \int p \log(p/q)\,\mathrm{d}x$. Then, for $q(x) = p(x \mid \theta + \mathrm{d}\theta)$, one expands

$$\log \frac{p(x \mid \theta)}{p(x \mid \theta + \mathrm{d}\theta)} = -\partial_i \log p\,\mathrm{d}\theta^i - \tfrac{1}{2}\,\partial_i\partial_j \log p\,\mathrm{d}\theta^i\,\mathrm{d}\theta^j + \mathcal{O}(\|\mathrm{d}\theta\|^3) \tag{8}$$

and integrates against $p(x \mid \theta)$. The linear term vanishes by score-centering, and the quadratic term yields $\frac{1}{2} g^{\mathrm{F}}_{ij}\mathrm{d}\theta^i\mathrm{d}\theta^j$. Full derivations are standard [Amari and Nagaoka, 2000, Cover and Thomas, 2006].

3.4 Why Fisher is canonical under coarse-graining

Under stochastic maps (Markov kernels) $p \mapsto p'$ that represent coarse-graining, distinguishability cannot increase. Chentsov's theorem states that Fisher is (up to scale) the unique Riemannian metric on the space of probability distributions satisfying monotonicity under all Markov morphisms [Chentsov, 1982, Amari, 2016]. For VDM, this provides a principled candidate metric when a coarse-graining map Π from microscopic states to statistical states is declared: the Fisher metric is the unique local metric compatible with the "information can only be lost" principle.

3.5 Worked example: univariate Gaussian family

For $p(x \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\big(-\frac{(x-\mu)^2}{2\sigma^2}\big)$, one computes

$$\partial_\mu \log p = \frac{x-\mu}{\sigma^2}, \qquad \partial_\sigma \log p = -\frac{1}{\sigma} + \frac{(x-\mu)^2}{\sigma^3}. \tag{9}$$

Taking expectations gives the Fisher matrix

$$g^{\mathrm{F}}(\mu,\sigma) = \begin{pmatrix} \sigma^{-2} & 0 \\ 0 & 2\,\sigma^{-2} \end{pmatrix} \tag{10}$$

in (μ, σ) coordinates, i.e. $g_{\mu\mu} = \sigma^{-2}$, $g_{\sigma\sigma} = 2\,\sigma^{-2}$, and $g_{\mu\sigma} = 0$. If one instead uses $\eta = \log\sigma$, then $g_{\eta\eta} = 2$ by the chain rule.

4 Cramér–Rao bound and operational meaning

4.1 Statement

Let $\hat{\theta}(x)$ be an unbiased estimator of θ based on n i.i.d. samples from $p(x \mid \theta)$. Then the Cramér–Rao inequality states

$$\mathrm{Cov}(\hat{\theta}) \succeq \frac{1}{n}\left(g^{\mathrm{F}}(\theta)\right)^{-1}, \tag{11}$$

where $A \succeq B$ denotes that $A - B$ is positive semidefinite [Cramér, 1946, Rao, 1945, Fisher, 1925].

4.2 Proof sketch (score/Cauchy–Schwarz)

Write the score for n samples as $s_i = \sum_{k=1}^{n} \partial_i \log p(x_k \mid \theta)$. Unbiasedness implies $\partial_i \mathbb{E}[\hat{\theta}^j] = \delta_i^j$. One shows $\mathrm{Cov}(\hat{\theta}, s) = I$ and then applies Cauchy–Schwarz to the random vectors $\hat{\theta} - \theta$ and s. The covariance of s is ng^{F}. Rearranging yields the bound.

4.3 Interpretation

The Fisher metric sets the best-possible local precision achievable by any unbiased estimator. For VDM, this becomes a concrete design constraint: if a coarse-grained coordinate θ is meant to represent an experimentally accessible observable, the Fisher metric quantifies the irreducible uncertainty imposed by the chosen observation protocol.

5 Thermodynamic geometry: Weinhold and Ruppeiner

5.1 State space and entropy concavity

In equilibrium thermodynamics, a macroscopic state can be coordinatized by extensive variables $X = (U, V, N, \dots)$ or by other Legendre-related coordinates (e.g., (T, V, N)). Stability implies concavity of entropy in extensive variables, which makes the entropy Hessian negative semidefinite [Callen, 1985].

5.2 Weinhold metric

Weinhold introduced a Riemannian metric defined as the Hessian of internal energy in extensive coordinates [Weinhold, 1975]:

$$g^{\mathrm{W}}_{ij} := \frac{\partial^2 U}{\partial X^i \, \partial X^j}. \tag{12}$$

5.3 Ruppeiner metric

Ruppeiner's thermodynamic metric is defined as the negative Hessian of *dimensionless* entropy $s := S/k_B$ [Ruppeiner, 1979, 1995]:

$$g^{\mathrm{R}}_{ij} := -\frac{\partial^2 s}{\partial X^i \, \partial X^j}. \tag{13}$$

The dimensionless choice aligns directly with statistical/information geometry (entropy as log-volume).

The Weinhold and Ruppeiner metrics are conformally related by temperature (in appropriate coordinates):

$$g^{\mathrm{R}} = \frac{1}{T}\, g^{\mathrm{W}}. \tag{14}$$

5.4 Fluctuation meaning and curvature heuristics

Near an equilibrium state, fluctuation theory yields an approximate Gaussian distribution over ΔX:

$$\mathbb{P}(\Delta X) \propto \exp\Big(-\tfrac{1}{2}\, g^{\mathrm{R}}_{ij}\, \Delta X^i \Delta X^j \Big), \tag{15}$$

so the covariance is approximately $(g^{\mathrm{R}})^{-1}$ [Ruppeiner, 1995].

The scalar curvature R of g^{R} is often interpreted as a proxy for interaction strength: ideal gases are flat ($R = 0$), while interacting systems can have $|R|$ that grows large near critical points [Ruppeiner, 1995]. We treat any more specific scaling law as *model-dependent* and therefore gateable rather than assumed.

6 Fisher–Ruppeiner bridge in canonical ensembles

The most practical bridge for CFN validation is the exponential-family/canonical ensemble case. MaxEnt (Jaynes) states that, given constraints $\mathbb{E}[T_a] = \eta_a$, the distribution maximizing entropy takes the form [Jaynes, 1957a,b]

$$p(x \mid \theta) = \exp\big(\theta^a T_a(x) - \psi(\theta)\big), \tag{16}$$

where θ are Lagrange multipliers, T_a sufficient statistics, and ψ is the log-partition function.

6.1 Fisher metric as covariance in multiplier space

For exponential families,

$$\partial_a \log p = T_a - \partial_a \psi, \tag{17}$$

so

$$g^{\mathrm{F}}_{ab}(\theta) = \mathrm{Cov}_\theta(T_a, T_b) = \partial_a \partial_b \psi(\theta) \tag{18}$$

[Amari and Nagaoka, 2000]. In particular, for the canonical ensemble $p \propto e^{-\beta E}$ one has

$$g^{\mathrm{F}}_{\beta\beta} = \mathrm{Var}(E). \tag{19}$$

6.2 Ruppeiner metric as inverse covariance in expectation space

Let $\eta_a := \mathbb{E}_\theta[T_a]$ be expectation coordinates. The Legendre dual potential is the (dimensionless) entropy

$$s(\eta) = \theta^a \eta_a - \psi(\theta), \tag{20}$$

with θ viewed as a function of η. One then finds

$$g^{\mathrm{R}}_{ab}(\eta) = -\frac{\partial^2 s}{\partial \eta_a\, \partial \eta_b} = \big(g^{\mathrm{F}}(\theta)\big)^{-1}_{ab} \tag{21}$$

(up to coordinate conventions), exhibiting the duality between Fisher covariance geometry and entropy-Hessian geometry [Amari and Nagaoka, 2000, Ruppeiner, 1995].

6.3 One-dimensional canonical check

For $\theta = \beta$ and $\eta = U = \mathbb{E}[E]$, the duality can be expressed as

$$g^{\mathrm{F}}_{\beta\beta}\, g^{\mathrm{R}}_{UU} = 1 \tag{22}$$

when g^{R} is defined from dimensionless entropy $s = S/k_B$. Using standard fluctuation relations, one may also write

$$\mathrm{Var}(E) = k_B T^2 C_V, \qquad g^{\mathrm{R}}_{UU} = \frac{1}{k_B T^2 C_V}, \tag{23}$$

so the product is again unity. These relations are directly testable by Monte Carlo sampling (variance) and by finite-difference thermodynamic derivatives (heat capacity).

7 Quantum Fisher information and the QGT

VDM uses quantum-geometric inputs in CF01. For completeness, we record the standard information-geometric connection. Quantum Fisher information generalizes Fisher geometry to quantum states (density matrices) [Helstrom, 1976, Braunstein and Caves, 1994]. For pure states $|\psi(\theta)\rangle$, the quantum Fisher metric reduces (up to a conventional factor) to the Fubini–Study metric, which coincides with the symmetric (metric) part of the quantum geometric tensor [Provost and Vallée, 1980]. This provides a clean conceptual chain:

$$\text{QGT metric} \Rightarrow \text{quantum Fisher} \Rightarrow \text{classical Fisher} \Rightarrow \text{Ruppeiner (MaxEnt dual)}.$$

8 From information geometry to the metriplectic metric bracket

8.1 Natural gradient and metric-induced flows

Let θ parameterize a coarse-grained statistical manifold induced by a declared coarse-graining map Π from microscopic states. Given a scalar functional $\Sigma(\theta)$ (candidate entropy), the *natural gradient* flow is

$$\dot{\theta}^i = M^{ij}(\theta)\,\partial_j\Sigma(\theta), \qquad M^{ij}(\theta) := \left(g^{\mathrm{F}}(\theta)\right)^{-1}_{ij}, \tag{24}$$

which is invariant under reparameterization and uses the intrinsic geometry [Amari, 1998]. Entropy production is

$$\dot{\Sigma} = \partial_i\Sigma\, M^{ij}\,\partial_j\Sigma \geq 0. \tag{25}$$

8.2 A metriplectic-compatible bracket form

GENERIC/metriplectic evolution can be written abstractly as $\dot{x} = L\,\nabla E + M\,\nabla S$ with degeneracy constraints that prevent cross-driving [Morrison, 1986, Grmela and "Ottinger, 1997, "Ottinger and Grmela, 1997]. In a finite-dimensional coarse-grained coordinate θ, a corresponding symmetric bracket is

$$(f, g)_M := \partial_i f\, M^{ij}(\theta)\,\partial_j g. \tag{26}$$

Choosing $M^{ij} = (g^{\mathrm{F}})^{-1}_{ij}$ makes $(f, f)_M \geq 0$. To enforce invariants (degeneracy), one may project out the invariant gradient directions:

$$M = P^{\top}\,(g^{\mathrm{F}})^{-1}\,P, \qquad P := I - \frac{\nabla\mathcal{I}\,\nabla\mathcal{I}^{\top}}{\|\nabla\mathcal{I}\|^2}, \tag{27}$$

so that $M\nabla\mathcal{I} = 0$ while preserving symmetry and PSD. (Here $\mathcal{I}$ denotes the invariant functional in the VDM split; compare the projection construction used in CF01.)

8.3 What is and is not being claimed

This CF does *not* claim that every physical dissipation operator must literally equal an inverse Fisher metric. It claims that once a model chooses to interpret coarse-grained dissipation as information loss under a declared coarse-graining, Fisher/Ruppeiner metrics supply a canonical candidate class and a tight suite of falsifiable constraints.

9 Gate suite summary

For publication friendliness, we restate the geometry gates in one place:

- **G1 (Coordinate invariance):** curvature scalar (or other invariants) agree across coordinate charts.
- **G2 (Fisher–Ruppeiner bridge):** covariance-based Fisher and Hessian-based Ruppeiner computations match the predicted duality in canonical toy systems.
- **G3 (Curvature–response diagnostics):** geometry-linked predictors outperform baselines (or else the "geometry predicts response" claim is rejected).

10 Integration map

Downstream use in VDM. This CF becomes operational when a concrete VDM truncation identifies: (i) a coarse-graining map Π, (ii) a state coordinate θ or X, (iii) an entropy functional Σ to be increased, and (iv) invariants $\mathcal{I}$ to be protected. At that point, Fisher/Ruppeiner geometry gives a candidate M and predicts measurable response relations.

Relationship to other CF modules.

- CF01 supplies quantum-geometric inputs (QGT) that reduce to quantum/classical Fisher geometry.
- CF02 supplies contact/geometric thermodynamics structure that pairs naturally with Ruppeiner/Weinhold metrics.

- CF04 uses Fisher-information concepts in causality-constrained transport modules; CF06 provides the metric foundations.
- CF05 provides closure discipline so that metric-driven relaxation is not silently constrained by unintended invariants.

11 Provenance and reproducibility

Commit: `91332f4cf89a52d239eaab6659b8df8d8b9f3202` **Seed(s):** `N/A` **Artifacts:** `Source: Derivation/Complete-Formalisms/CF06_Info_Geom_Fisher_Ruppeiner_Foundations.md (SHA256: 4bcec68906c71a8034cdd9a4143ae3811c35dac1a8122580969e26910a50cd00)`

12 Limitations and future work

- **Nonequilibrium extensions:** Ruppeiner geometry is equilibrium-based. Nonequilibrium thermodynamic metrics require explicit modeling (e.g., extended state variables or GENERIC closures).
- **Curvature interpretations are model-dependent:** "interaction strength" and "correlation volume" heuristics must be validated per model; we intentionally gate rather than assume.
- **Quantum mixed states:** for mixed states there are multiple monotone quantum metrics; choosing one is a physical modeling decision beyond the pure-state QGT equivalence.
- **Coarse-graining choice dominates:** identifying Π is the true physical input. Fisher geometry is canonical *given* Π, not a substitute for specifying it.

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